



OIBSIP • DATA ANALYTICS INTERNSHIP • PROJECT 3 / LEVEL 2

Fraud Detection

Using Machine Learning

Detecting fraudulent credit card transactions through predictive analytics, ensemble learning, and real-time risk scoring.

Ashfiya

Aspiring Data Analyst | ML Enthusiast

Project Overview

Fraudulent financial transactions cause significant losses to businesses and financial institutions every year.

This project applies Machine Learning to identify potentially fraudulent credit card transactions and strengthen transaction security.

The model analyzes transaction patterns and classifies each transaction as legitimate or fraudulent — helping organizations flag suspicious activity in real time.

WHY IT MATTERS

0.17%

of all transactions are fraudulent — yet they drive massive financial losses industry-wide.

GOAL

Classify each transaction as Legitimate or Fraudulent with high precision and recall.



Detect Fraud



Analyze Patterns



Build ML Model



Evaluate Results

Objectives

What this project sets out to achieve



01

Detect Fraud

Identify fraudulent credit card transactions accurately and consistently.



02

Analyze Behavior

Study transaction behavior patterns to surface hidden fraud signals.



03

Build the Model

Construct a robust machine learning classification model.



04

Evaluate Rigorously

Assess performance using industry-standard evaluation metrics.



05

Improve Security

Strengthen financial security through predictive analytics.

The Dataset

Anonymized credit card transaction records used to train and evaluate the fraud classifier.

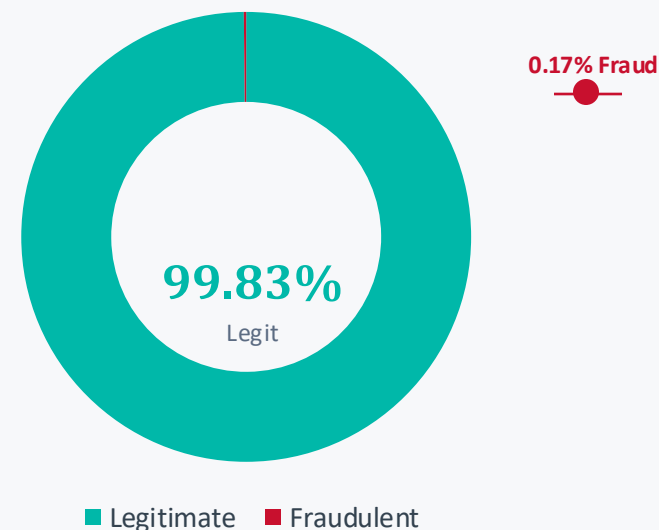
- Transaction-level numerical features (PCA-anonymized)
- Transaction amount
- Transaction time
- Customer behavior indicators
- Fraud classification label

TARGET VARIABLE — Class

0 = Legitimate Transaction

1 = Fraudulent Transaction

CLASS DISTRIBUTION (Typical)



Severe class imbalance is the defining challenge of fraud detection — addressed later via SMOTE.

Technologies Used

The tech stack behind the fraud detection pipeline



Programming Language

Python



Data Analysis

Pandas, NumPy



Data Visualization

**Matplotlib,
Seaborn**



Machine Learning

Scikit-Learn

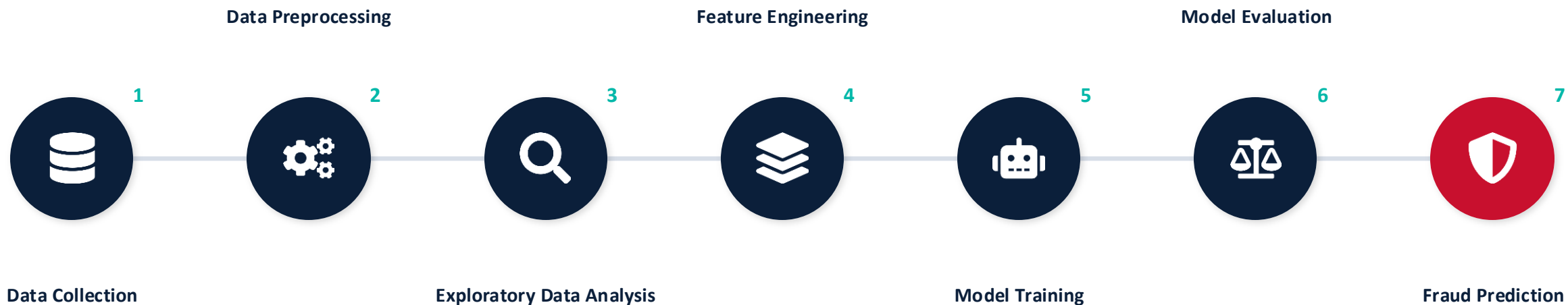


Development Environment

Jupyter Notebook

Project Workflow

End-to-end pipeline from raw data to fraud prediction



A self-paced, end-to-end machine learning pipeline — from raw transactions to actionable fraud predictions.

Exploratory Data Analysis

Understanding the data before modeling



Transaction Distribution

Examined volume and spread of transaction amounts across the dataset.



Fraud vs Non-Fraud

Compared behavioral patterns between legitimate and fraudulent transactions.



Feature Correlations

Mapped relationships between anonymized features using correlation heatmaps.



Outlier Detection

Identified extreme values that may signal anomalous, high-risk activity.



Data Imbalance

Quantified the fraud-to-legitimate ratio to inform resampling strategy.

Machine Learning Model



Random Forest

Classifier

An ensemble of decision trees that vote together — reducing overfitting and capturing complex, non-linear fraud patterns that a single model would miss.

WHY RANDOM FOREST?



Handles large datasets efficiently



Reduces overfitting through ensemble learning



Works well with imbalanced datasets



Provides feature importance analysis



Delivers high classification accuracy

Model Performance

Evaluated using industry-standard classification metrics

- Accuracy Score
- Precision
- Recall
- F1-Score
- Confusion Matrix

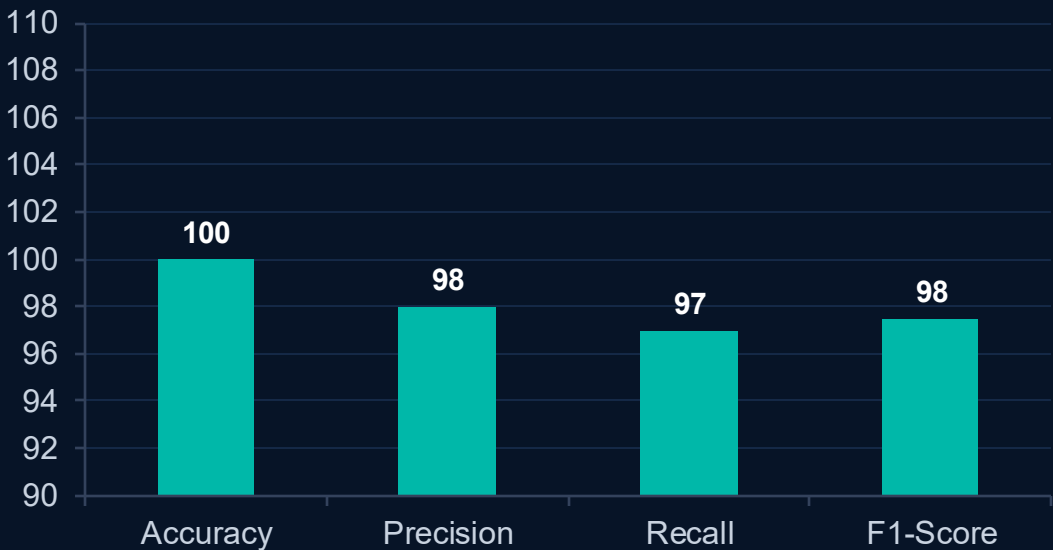
ACCURACY ACHIEVED

100%

Model TypeRandom Forest Classifier

ClassificationFraud / Non-Fraud

CLASSIFICATION METRICS



Repository Structure

How the project is organized on GitHub



OIBSIP_Fraud_Detection/

- fraud.ipynb
- Fraud_Detection.html
- creditcardfraud.htm
- README.md

- Dataset Overview.png
- Correlation Heatmap.png
- Confusion Matrix.png
- Load Dataset.png

- Presentation.pdf



Notebooks & Reports

fraud.ipynb, Fraud_Detection.html, Presentation.pdf



Visual Outputs

Dataset Overview, Correlation Heatmap, Confusion Matrix, Load Dataset



Documentation

README.md with full project write-up

Future Improvements

Where this project goes from here



Deploy via Streamlit

Make the model accessible through an interactive web app.



Real-time Detection

Add live fraud-detection capability for streaming transactions.



Model Comparison

Benchmark against additional ML algorithms for best performance.



Hyperparameter Tuning

Fine-tune model parameters to push performance further.



Interactive Dashboard

Build a dashboard for live monitoring and reporting.



Cloud Deployment

Integrate with cloud infrastructure for production scale.

Business Impact

How machine learning translates into organizational value



Reduce Financial Losses

Minimizes the financial impact of undetected fraudulent transactions.



Improve Monitoring

Strengthens real-time transaction monitoring capabilities.



Enhance Customer Trust

Builds confidence by proactively protecting customer accounts.



Support Risk Management

Equips risk teams with predictive, data-driven decision tools.

Key Skills Demonstrated

Capabilities applied throughout this project



Data Cleaning



Exploratory Data Analysis (EDA)



Data Visualization



Machine Learning



Classification Modeling



Performance Evaluation



Fraud Analytics



Ashfiya

Aspiring Data Analyst | Machine Learning Enthusiast

OIBSIP Data Analytics Internship — Project 3, Level 2



Thank You

Questions & Discussion Welcome

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